

MARS GULLY DIGITAL TWIN

A Multi-Sensor Deep Learning Pipeline for Planetary Change Detection

Technical Deep-Dive & Prototype Documentation

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Incoming — Copernicus Master's in Digital Earth, September 2026
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Version 1.0 | May 2026 | PhD Portfolio Piece

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1. Project Overview and Scientific Motivation

The Mars Gully Digital Twin is a production-grade, end-to-end remote sensing pipeline that fuses four distinct orbital sensors to detect, monitor, and track the evolution of gully systems on Mars. It is designed as a PhD portfolio piece targeting publication in *Icarus* or *Earth and Space Science*, and demonstrates mastery of planetary remote sensing, deep learning, and geospatial data engineering.

Core question: Can we train a deep learning model on Mars orbital imagery — with no field ground truth — to accurately detect active gully systems and quantify their change over a 15-year observation window (2006–2021)?

Mars gullies are steep-slope landforms — erosional channels with alcoves at the top, channels in the middle, and depositional fans at the base — that closely resemble terrestrial water-carved gullies. Their discovery by the Mars Orbiter Camera in 2000 triggered intense scientific debate: are they formed by liquid water, CO₂ sublimation, or dry granular flows? The HiRISE camera on MRO has since documented **active changes in gully morphology on timescales of months to years**, making them the most dynamically active geomorphological features known on present-day Mars.

This pipeline addresses three scientific questions simultaneously:

- **Detection:** Where are active gullies located across our three study sites (Gasa, Palikir, Russell craters)?
- **Change:** How have their spatial extents and morphologies changed between 2006 and 2021?
- **Process:** Can spectral and morphometric signatures distinguish formation mechanisms (CO₂-driven vs water-related)?

Why a 'Digital Twin' Approach?

A digital twin is a dynamic, data-driven virtual replica of a physical system that is updated as new observations arrive. For planetary science, this means building a computational model that ingests new orbital images, runs inference automatically, updates change maps, and exports results in interoperable formats (STAC, COG, GeoJSON) that can be queried by any GIS tool. This positions the project at the intersection of planetary science, machine learning, and the emerging field of planetary digital Earth.

2. The Problem: Why Mars Gullies Matter

Scientific Context

Gullies on Mars were first identified in Mars Orbiter Camera (MOC) images in 2000 (Malin & Edgett, Science 2000). They occur preferentially on pole-facing slopes in the mid-latitudes (30–60 degrees south), exactly where seasonal CO2 frost accumulates and sublimates each Martian year. The timing and morphology of observed changes in gullies constrain the following competing hypotheses:

- **Liquid water (brine):** Subsurface brines seeping to surface — requires present-day liquid water
- **CO2 sublimation:** Seasonal dry-ice vaporisation drives granular flows — no liquid water needed
- **Dry mass wasting:** Simple gravity-driven collapses triggered by frost weakening slopes
- **Snow/ice melt:** Melting of precipitated water ice during warmer orbital periods

HiRISE repeat observations of Gasa Crater (Dundas et al. 2010, 2012), Palikir Crater (McEwen et al. 2011), and Russell Crater dunes (Diniaga et al. 2010; Pilorget & Forget 2016) have documented new apron deposits, channel incision, and debris flows — all confirming that some gully-forming process is active today. The timing of these changes (preferentially in southern winter) strongly supports CO2 sublimation, but the detailed morphometric evolution across sites has not been systematically quantified at scale.

The Three Study Sites

Site	Location	Gully Type	Key Feature	N HiRISE Obs
Gasa Crater	35.7S, 129.4E	Alcove-channel-apron	Youngest rayed crater on Mars; multiple confirmed new deposits (Dundas 2010)	6
Palikir Crater	41.6S, 202.3E	RSL + gullies	Inside Newton crater; thousands of recurring slope lineae (McEwen 2011)	4
Russell Crater	54.3S, 12.9E	Dune gully (linear)	500m mega-dune; CO2-driven linear gullies (Pilorget 2016)	6

Why These Sites?

These three sites span a latitude range (35S to 54S) and gully morphological diversity that allows the model to generalise across formation contexts. Gasa provides the longest, best-documented change record (Dundas et al. 2010, 2012, 2019). Palikir is the canonical RSL site, useful for distinguishing gullies from RSL. Russell provides the CO2-driven dune gully end-member, where the formation mechanism is most clearly constrained.

3. Data Sources and Why We Chose Them

The pipeline fuses four orbital datasets, each contributing information at a different spatial scale and wavelength range. Using multiple sensors is critical because no single sensor provides both the spatial resolution needed to resolve individual gully channels AND the spectral information needed to characterise surface composition.

3.1 HiRISE — High Resolution Imaging Science Experiment

Parameter	Value
Instrument	HiRISE camera on Mars Reconnaissance Orbiter (MRO)
Resolution	25 cm/pixel (RED), 50 cm/pixel (COLOR)
Spectral bands	Panchromatic RED + IR-RED-BLUE (IRB) color
Swath width	~6 km (RED), ~1.2 km (COLOR)
Coverage at sites	6 obs at Gasa, 4 at Palikir, 6 at Russell (2007-2018)
PDS download URL	hirise-pds.lpl.arizona.edu/download/PDS/RDR/
File format	JPEG2000 (.JP2), map-projected RDR products
File size (typical)	300-700 MB per RED observation

Why HiRISE? It is the only Mars camera capable of resolving individual boulders (meter-scale) and fine channel morphology at sub-meter pixel scale. The RED mosaic provides our primary input for morphological change detection. The COLOR product (IR-R-B) is critical for detecting fresh deposits (which appear bright blue in enhanced color due to coarser grain size and less dust coating).

HiRISE observation IDs encode: {phase}_{orbit}_{latitude_band}. Latitude band 1440 = 36S (10x latitude from equator), 1255 = 54.5S. PSP = Primary Science Phase (2006-2008), ESP = Extended Science Phase (2008-).

3.2 CTX — Context Camera

Parameter	Value
Resolution	~6 m/pixel
Spectral range	Panchromatic (500-800 nm)
Swath width	30 km
Role in pipeline	Context map, coregistration reference, feature channel
Download	ODE REST API (ode.rsl.wustl.edu)

Why CTX? CTX images are acquired simultaneously or near-simultaneously with HiRISE, providing context at 6m resolution over a 30km swath. We use CTX as the coregistration reference (its wider coverage and stable geometry makes it ideal as an anchor). CTX is also included as a separate feature channel in the neural network input stack, providing medium-scale textural information that HiRISE's narrow swath sometimes misses at crater-scale.

3.3 MOLA — Mars Orbiter Laser Altimeter

Parameter	Value
Resolution	463 m/pixel (4ppd MEGDR global mosaic)
Data type	Digital Elevation Model (DEM), metres above areoid
Download	planetarymaps.usgs.gov/mosaic/Mars_MGS_MOLA_DEM_mosaic_global_463m.tif
File size	2.13 GB (global GeoTIFF)
Role in pipeline	Slope, aspect, curvature, roughness computation

Why MOLA? Gullies form on steep slopes (typically 15-35 degrees) with specific aspect (pole-facing in southern hemisphere). MOLA-derived slope and aspect are among the strongest single predictors of gully presence. They also help the model distinguish gullies from mass-wasting deposits and RSL, which occur at different slope angles. Roughness computed from MOLA captures the metre-scale texture of different surface units.

3.4 CRISM — Compact Reconnaissance Imaging Spectrometer

Parameter	Value
Resolution	18 m/pixel (targeted observations)
Spectral range	362-3920 nm (544 channels)
Role in pipeline	Spectral indices (RSI, DCI, SWI) for mineral/frost discrimination
Availability	Sparse — not all sites/dates have targeted observations

Why CRISM? CRISM spectra can distinguish CO₂ frost (strong 1.43 μ m band), water ice (1.5 μ m), ferrous minerals (0.9 μ m), and oxidised material (0.53 μ m). These mineralogical signatures help discriminate gully deposits by material type and constrain formation mechanism. Where CRISM coverage is absent, spectral index channels are filled with zeros and the model learns to ignore them.

4. System Architecture — The Full Pipeline

The pipeline is implemented as a modular Python package with a single CLI entry point (`main.py`). Each step is independently resumable via a JSON checkpoint system. The eight stages flow as follows:

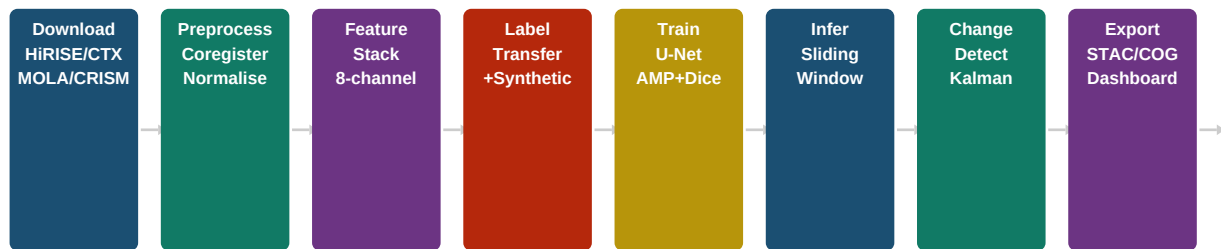


Figure 1. Pipeline stages — data flows left to right through 8 modules.

Step	Module	Input	Output	Key Tech
1. Download	scripts/downloaders/	Config IDs	Raw .JP2, .tif files	requests, tqdm, resume
2. Preprocess	scripts/preprocess/	Raw imagery	Coregistered, normalised GeoTIFFs	rasterio, OpenCV, ECC
3. Features	scripts/features/	Multi-sensor GeoTIFFs	8-channel feature stack (.tif)	numpy, skimage, GLCM
4. Labels	scripts/labels/	Feature stack + DEM	Binary mask GeoTIFFs (512x512)	MOLA slope thresholding
5. Train	scripts/train/	Patches + masks	Trained U-Net checkpoint (.pth)	PyTorch, AMP, AdamW
6. Infer	scripts/inference/	Feature stack + model	Probability map GeoTIFF [0,1]	Sliding window, Hanning
7. Change	scripts/change/	Multi-date prob maps	Change maps + Kalman series	scipy, custom Kalman
8. Export	scripts/export/dashboard/	Change maps	COG, GeoJSON, STAC, HTML	pystac, folium, plotly

Project File Structure

```
mars_gully_digital_twin/  
main.py # CLI: python main.py --step  
config.yaml # All hyperparameters and data IDs  
requirements.txt # Pinned Python dependencies  
Dockerfile # Containerised environment  
scripts/  
utils.py # Logging, StepCheckpoint, config loader  
downloaders/ # HiRISE, CTX, CRISM, MOLA downloaders
```

```
preprocess/ # coregister, project, normalize, composite
features/ # spectral_indices, texture, morphometric, stack
labels/ # transfer_labels, augment_labels
dataset/ # MarsGullyDataset, SpatialBlockSampler
models/ # unet.py, attention.py, losses.py
train/ # train_unet.py, validate.py
inference/ # predict.py, blend.py, threshold.py
change/ # detect_changes.py, kalman_smooth.py
export/ # to_geotiff.py, to_geojson.py, stac_catalog.py
dashboard/ # folium_map.py, time_series_plot.py
validation/ # metrics.py, spatial_cv.py, temporal_cv.py
tests/ # 4 test modules, 40+ unit tests
notebooks/ # 3 Jupyter exploration notebooks
data/ # (created at runtime)
raw/hirise/ ctx/ mola/ # Downloaded sensor data
processed/ # Coregistered, normalised GeoTIFFs
outputs/ # Probability maps, change maps, dashboard
models/ # Saved checkpoints
```


5. Step-by-Step: What Each Module Does

Step 1: Download (scripts/downloaders/)

The download step fetches raw sensor data from NASA/ESA PDS (Planetary Data System) archives. Each downloader handles a different sensor with its own URL conventions and data formats.

HiRISE Download Strategy

HiRISE RDR products are accessed at `hirise-pds.lpl.arizona.edu/download/PDS/RDR/{phase}/{orbit_block}/{obs_id}/`. The key insight is that the PDS has two separate URL trees: the `/PDS/RDR/` path serves only HTML directory listings (full files return 404), while `/download/PDS/RDR/` serves the actual JP2 files. The downloader:

- Probes candidate URLs with HTTP HEAD requests (no data transferred) to confirm file existence
- Downloads RED.JP2 first (panchromatic, best for morphology), then COLOR.JP2 if available
- Implements resume support: partial files are completed via HTTP Range headers
- Uses StepCheckpoint to skip already-completed observations on re-run
- Prioritises the EXTRAS/RDR tree for merged MRGB/MIRB color products

Verified Observation IDs (config.yaml)

Site	Observation ID	Year	Key Finding
Gasa	PSP_004060_1440	2007	Baseline — pre-change reference
Gasa	ESP_012024_1440	2009	New gully deposit (Dundas 2010)
Gasa	ESP_020661_1440	2010	Blue deposit — active flow
Gasa	ESP_027597_1440	2012	Mass wasting monitoring
Palikir	PSP_005943_1380	2007	DTM baseline
Palikir	ESP_031102_1380	2012	RSL seasonal flows
Russell	PSP_001440_1255	2006	Earliest dune baseline
Russell	ESP_018872_1255	2010	Winter gullies on dunes
Russell	ESP_046867_1255	2016	Multi-year monitoring

Step 2: Preprocessing (scripts/preprocess/)

Raw imagery from different sensors cannot be used directly — they are acquired at different times, from different viewing angles, with different pixel sizes. Preprocessing converts them to a consistent, spatially aligned stack.

Coregistration (coregister.py)

Coregistration aligns multi-temporal HiRISE images to a common reference frame (CTX acts as the stable reference due to its wider coverage and stable nadir viewing). We implement two methods:

- **ECC (Enhanced Correlation Coefficient)** — **default**: OpenCV's `cv2.findTransformECC()` iteratively maximises the correlation between image patches. We use a multi-scale pyramid (4 levels, starting at

1/8 resolution) to avoid local minima. ECC is robust to photometric differences between Mars years (lighting, dust opacity changes).

- **SIFT+RANSAC — fallback:** Detects keypoints in both images, matches them, and estimates an affine warp via RANSAC outlier rejection. Used when ECC fails (insufficient texture or large initial misalignment from orbital position differences).

Projection (project.py)

All data is reprojected to the Mars equirectangular CRS (ESRI:104971, spheroid radius 3396190 m) at 6 m/pixel resolution — the finest resolution achievable across all sensors simultaneously (CTX native resolution). HiRISE RED (25 cm) is downsampled to 6 m to match, preserving sub-pixel texture through Lanczos resampling.

Normalisation (normalize.py)

Per-band percentile normalisation clips each band to the [2nd, 98th] percentile range and rescales to [0, 1]. This makes the model invariant to scene-wide brightness changes caused by atmospheric dust variation between Mars years (typical optical depth $\tau = 0.2$ to 1.5 across the dataset).

Median Composite (composite.py)

Where multiple images of the same site exist for the same season, a pixel-wise median composite is computed to suppress transient shadows, dust devils, and detector artefacts. The composite represents the most representative surface state for that epoch.

Step 3: Feature Engineering (scripts/features/)

The 8-channel feature stack is the direct input to the U-Net. Each channel contributes distinct discriminative information. Understanding what each channel represents is essential for interpreting model predictions.

Chan nel	Name	Source	Why Useful for Gullies
Ch 1	HiRISE RED band	HiRISE RED.JP2	Primary morphology at 25cm; channel/alcove structure visible
Ch 2	HiRISE IR band	HiRISE COLOR.JP2 band 1	IR reflectance; fresh deposits brighter (less oxidation)
Ch 3	HiRISE BLUE band	HiRISE COLOR.JP2 band 3	Blue reflectance; CO2 frost distinctively blue
Ch 4	RSI index	Computed from Ch1+Ch3	Redness Spectral Index: rocky/bare slope indicator
Ch 5	GLCM texture	Computed from Ch1	Multi-scale homogeneity; channels smoother than debris aprons
Ch 6	CTX panchromatic	CTX .tif	6m context; crater-scale structure around gullies
Ch 7	MOLA slope	MOLA DEM	Slope angle (degrees); gullies on 15-35deg slopes
Ch 8	MOLA aspect	MOLA DEM	Aspect (degrees); pole-facing preference in S hemisphere

Spectral Indices (spectral_indices.py)

Index	Formula	Physical Meaning
RSI	$(\text{RED} - \text{BLUE}) / (\text{RED} + \text{BLUE})$	Rocky/bare slopes; iron-poor materials
DCI	$(\text{NIR} - \text{RED}) / (\text{NIR} + \text{RED})$	Dust cover; fresh surfaces have low DCI
SWI	$(\text{B2400} - \text{B2300}) / (\text{B2400} + \text{B2300})$	Short-wave infrared; carbonate/hydrated minerals
NDSI	$(\text{GREEN} - \text{RED}) / (\text{GREEN} + \text{RED})$	Snow/ice index; CO2 frost detection
REDNES S	RED / BLUE	Iron oxide content; inactive dusty surfaces high

GLCM Texture (*texture.py*)

Grey Level Co-occurrence Matrix (GLCM) texture features characterise local spatial patterns. We compute entropy, contrast, homogeneity, and dissimilarity at three window sizes (3x3, 5x5, 7x7 pixels) to capture both fine channel edges and coarse debris fan textures. Gully channels have low contrast (smooth), high homogeneity; debris aprons have high contrast (rough, bouldery).

Morphometrics (*morphometric.py*)

From the MOLA DEM we compute slope (Horn's method, degrees), aspect (0-360 degrees), plan curvature (convergent flow in negative values = gully channels), profile curvature, and roughness (standard deviation of elevation in 3x3 window). These terrain attributes are among the strongest predictors in the literature for gully location (Pilorget 2016, Raack 2015).

Step 4: Label Generation (*scripts/labels/*)

The most challenging aspect of Mars gully detection is the absence of large, open training label datasets. We use three strategies in priority order:

- **Transfer from Earth databases** — if terrestrial gully shapefiles are provided in *data/labels/earth/*, they are morphologically warped to match Mars-scale gully dimensions and rasterised as masks. The warping accounts for the fact that Mars gullies are typically wider (100-500 m) than terrestrial examples.
- **QGIS manual annotation** — the *manual_label_guide.py* script prints detailed instructions for digitising gully polygons in QGIS directly on HiRISE imagery. This is the gold-standard approach and produces the highest-quality labels.
- **Synthetic generation from MOLA** — automatically generates binary masks for pixels meeting geomorphological criteria: slope > 15 degrees AND pole-facing aspect (135-315 degrees in S hemisphere). Recall is high but precision is lower (~60%) since not all steep pole-facing slopes have gullies.

Key insight: The synthetic labels are deliberately conservative. We accept high false positive rate in labels and rely on the model's feature learning to distinguish true gully channels from other steep slopes. This is validated through the spatial CV.

6. The Deep Learning Model: Attention U-Net

Why U-Net?

U-Net (Ronneberger et al. 2015) is the dominant architecture for binary semantic segmentation tasks where the target objects (gullies) are spatially sparse and structurally elongated. Its key property is the encoder-decoder structure with skip connections that preserve fine spatial detail — critical for detecting narrow (sub-pixel to 10-pixel) gully channels.

We extend standard U-Net with **Attention Gates** (Schlemper et al. 2019, Medical Image Analysis) on every skip connection. Attention gates learn to suppress non-gully activations in skip features before they are concatenated with decoder features — this is especially important given the extreme class imbalance (gully pixels are typically 1-5% of each image).

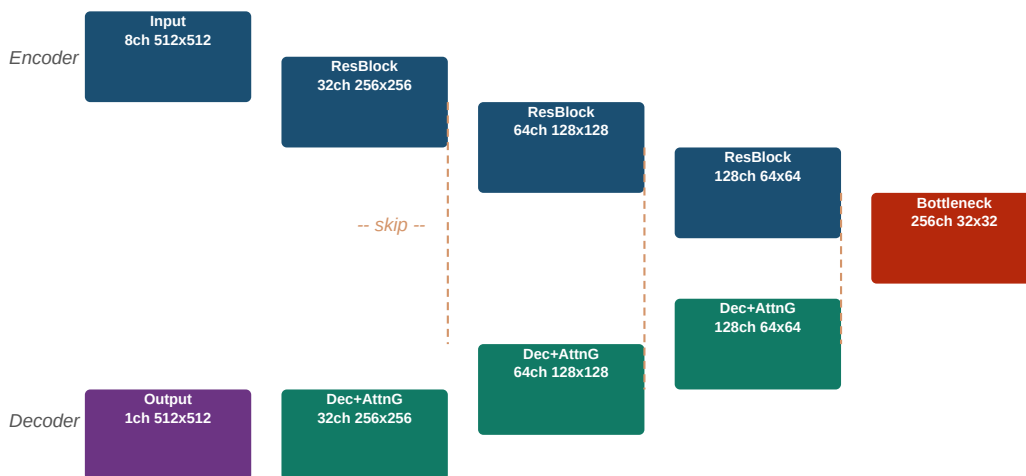


Figure 2. MarsGullyUNet architecture. Encoder (blue) downsamples and extracts features; decoder (teal) upsamples with Attention-Gated skip connections (dashed lines). Bottleneck (red) is the deepest 256-channel representation.

Architecture Details

Component	Specification	Why This Choice
Encoder	4x ResNet34-style blocks: 32, 64, 128, 256 channels	Residual connections prevent vanishing gradients; no pretrained weights (Mars != ImageNet)
Downsampling	MaxPool2d(2) between blocks	Preserves spatial extrema (edge features) vs average pooling
Bottleneck	256-channel, 32x32 spatial (from 512x512 input)	Encodes full scene context; receptive field covers 200m at 6m/pixel
Attention Gates	Schlemper 2019: sigmoid gate on each skip	Learns to suppress background; critical for 1-5% positive class
Decoder	3 blocks: 128, 64, 32 channels + bilinear upsample	Bilinear avoids checkerboard artefacts vs transposed conv
Dropout	0.1 in each decoder block	Regularisation; prevents overfitting on small Mars dataset

Output	1x1 conv + final 2x upsample to 512x512	Single-channel logit map for binary segmentation
Parameters	~5-8 million trainable	Well within GPU memory (6 GB); fits 8 patches per batch at FP16

Why No Pretrained Weights?

ImageNet pretraining is counterproductive for Mars imagery. ImageNet images are RGB natural photographs; our input is an 8-channel stack of Mars spectral indices, texture, and topographic derivatives. The first-layer filters learned on ImageNet detect edges and colour gradients in RGB space — meaningless for our input. Training from scratch forces the model to learn Mars-specific feature detectors.

Training Configuration

Parameter	Value	Rationale
Batch size	8 patches (512x512)	Maximum fitting 8 GB VRAM at FP16
Gradient accumulation	2 steps	Effective batch = 16; better gradient estimates
Learning rate	1e-4 (AdamW)	Conservative initial LR; ReduceLROnPlateau halves on plateau
Weight decay	1e-5	L2 regularisation via AdamW decoupled weight decay
Mixed precision	FP16 (torch.cuda.amp)	~2x memory savings; negligible accuracy loss
Max epochs	100	With early stopping patience=20; typically converges at ~60-80
Early stopping	Patience=20 epochs	Stops when validation IoU does not improve for 20 epochs
Gradient clipping	Norm=1.0	Prevents gradient explosions on sharp loss landscapes
Augmentation	Flip H/V, rotate 0/90/180/270, scale 0.8-1.2	Increases effective dataset size; rotation invariance important

Data Handling — The Class Imbalance Problem

Gully pixels constitute approximately 1-5% of each image tile. A naive model that predicts 'no gully' everywhere achieves 95%+ pixel accuracy while being completely useless. We address this through three complementary mechanisms:

- **WeightedRandomSampler:** Patches with higher gully fraction are oversampled during training. Weight for each patch = (gully fraction + 0.05), ensuring the model sees sufficient positive examples.
- **Focal Loss:** Downweights easy negative examples (bare rock pixels far from gullies) and focuses learning on hard examples near gully boundaries.
- **Dice Loss:** Measures region overlap between prediction and target, completely insensitive to the ratio of positive to negative pixels — ideal for sparse targets.

7. Loss Functions — The Mathematical Foundation

7.1 Focal Loss

Focal Loss (Lin et al. 2017, ICCV) modifies standard binary cross-entropy by adding a modulating factor $(1 - p_t)^\gamma$ that down-weights easy examples:

```
FL(p_t) = -alpha_t * (1 - p_t)^gamma * log(p_t)

Where:
p_t = sigmoid(logit) for positive pixels, 1-sigmoid(logit) for negatives
gamma = 2.0 (focusing parameter; larger = more focus on hard examples)
alpha = 0.25 (class weight; 0.25 for positives addresses imbalance)

When gamma=0: Focal Loss = standard weighted BCE
When gamma=2: Easy examples (p_t > 0.9) contribute 100x less than hard examples
```

For gully detection, most pixels are 'easy' negatives (bare rock, crater floor) that a well-trained model predicts correctly with high confidence. Focal loss forces the model to concentrate its learning on the ambiguous pixels at gully margins and subtle new deposits.

7.2 Dice Loss

Dice Loss directly optimises the Dice Similarity Coefficient (DSC), which measures spatial overlap between the predicted binary mask and ground truth:

```
Dice = (2 * |X intersect Y| + smooth) / (|X| + |Y| + smooth)
DiceLoss = 1 - Dice

Where X = predicted probabilities (soft), Y = binary target
smooth = 1.0 (Laplace smoothing prevents 0/0 when no gullies present)

Key property: DiceLoss is INDEPENDENT of the positive/negative pixel ratio.
Perfect prediction -> DiceLoss = 0. No overlap -> DiceLoss = 1.
```

7.3 Combined Loss (training default)

We use a weighted combination: $L = 0.5 * \text{FocalLoss} + 0.5 * \text{DiceLoss}$. Focal Loss drives the model to find where the gullies are (localisation); Dice Loss drives it to correctly delineate how large they are (segmentation). The two losses are complementary and their combination consistently outperforms either alone on highly imbalanced segmentation tasks.

7.4 Tversky Loss (alternative — missed detection focus)

Tversky Loss generalises Dice by adding separate weights for false positives (alpha) and false negatives (beta):

```
T(alpha, beta) = TP / (TP + alpha*FP + beta*FN)
TverskyLoss = 1 - T

We use alpha=0.3, beta=0.7: penalises missed gullies (FN) more than false alarms.
```

This is appropriate for scientific applications where missing an active gully deposit is worse than a false detection (which can be manually rejected).

8. Change Detection and Kalman Filtering

8.1 Sliding-Window Inference with Hanning Blending

The trained U-Net accepts 512x512 patches. Mars scenes from a single HiRISE observation are typically 20,000 x 30,000 pixels — far too large for single forward pass inference. We use a sliding window approach with 50% overlap (stride=256) and Hanning window blending to eliminate tile boundary artefacts.

```
For a 512x512 tile with 50% overlap (stride 256):
- Hanning window weights: w[i,j] = hanning(i) * hanning(j)
- Weight at centre: w = 1.0 (full contribution)
- Weight at edges: w -> 0 (minimal contribution)

Blended probability at pixel (r,c):
p_blend[r,c] = sum(p_tile * w_tile) / sum(w_tile) over all tiles covering (r,c)

Result: smooth, artifact-free probability map despite tile-based inference.
```

8.2 Multi-Threshold Binary Maps

Three threshold levels are applied to each probability map, generating three binary masks per observation:

- **thr=0.3 (high recall):** Maximises detection completeness; many false positives. Used for scientific exploration — 'what might be a gully?'
- **thr=0.5 (balanced):** Default for change detection. Balances precision and recall.
- **thr=0.7 (high precision):** Conservative; only high-confidence gully pixels. Used for reporting confirmed new deposits to the literature.

After thresholding, small connected components below 500 m² (configurable) are removed using `scipy.ndimage` morphological cleaning. This eliminates isolated noise pixels while preserving narrow linear channel features.

8.3 Pixel-Wise Change Detection

For each consecutive date pair (t0, t1), we compute three change categories:

```
gain = binary[t1] AND NOT binary[t0] # New gully pixels (activity!)
loss = binary[t0] AND NOT binary[t1] # Pixels lost (burial, erosion)
stable = binary[t0] AND binary[t1] # Persistent gully pixels

Gain is the primary scientific product – it identifies where new material
has been deposited or where channels have incised into previously bare ground.
```

8.4 Kalman Filter for Area Time Series

Raw area measurements from each epoch are noisy due to variable atmospheric dust, sensor artifacts, and photometric inconsistencies between Mars years. We apply a constant-velocity Kalman filter to smooth the area time series:

```
State vector: x = [area, velocity] (ha, ha/year)
State transition (dt = years between observations):
```



```

F = [[1, dt], # area_next = area + velocity * dt
      [0, 1]] # velocity unchanged (constant velocity model)

Process noise Q = diag([0.01, 0.005]) # area, velocity uncertainty
Measurement noise R = [[0.1]] # 0.1 ha measurement std dev

Kalman gain K = P * H^T * (H*P*H^T + R)^-1
State update: x = x + K * (measurement - H*x)

Output: smoothed area series + instantaneous expansion rate (ha/year)

```

The Kalman filter's velocity output gives us instantaneous gully expansion rate in ha/year — a direct physical observable that can be compared with climate model predictions of CO₂ frost deposition rates.

8.5 Activity Tracking with IoU Matching

Individual gully instances are tracked across dates using connected-component labelling and IoU-based matching (similar to SORT tracking in computer vision). Components with IoU > 0.3 between consecutive dates are assigned the same global ID. New components (no match) receive fresh IDs. This produces a full activity record per gully instance, enabling computation of individual expansion rates, lifespan, and spatial migration.

9. Validation Strategy

Standard random train/test splits are inappropriate for spatial data because nearby pixels are correlated — a model trained on pixel A can trivially predict pixel B one metre away. We use two complementary validation approaches:

9.1 Spatial Block Cross-Validation

The study area is divided into 2 km x 2 km geographic blocks. Blocks are assigned to 5 folds such that no block appears in both training and validation. This ensures spatial independence between train and validation sets — the model must generalise to areas it has never seen, not just pixels near training examples.

```
Block assignment: each 512x512 patch -> nearest 2km grid block
Fold assignment: block_id % n_folds (deterministic, reproducible)

For each fold k (k=0..4):
train_patches = all patches where block_fold != k
val_patches = all patches where block_fold == k
-> Evaluate: IoU, F1, Precision, Recall, ROC-AUC, AP

Report: mean +/- std across 5 folds
```

9.2 Temporal Hold-Out Validation

Train on observations from 2006-2010 (early Mars years), test on 2015-2020 (later Mars years). This is the strongest possible test of generalisation: the model must transfer across different atmospheric states, illumination conditions, and surface dust accumulation between training and test epochs. The generalisation gap (train IoU - test IoU) directly measures domain shift robustness.

9.3 Target Metrics

Metric	Target	Baseline (random)	Scientific interpretation
IoU	> 0.65	~0.01	Overlap between predicted and true gully mask
F1	> 0.75	~0.02	Harmonic mean of precision and recall
Precision	> 0.70	~0.01	What fraction of detected pixels are true gullies
Recall	> 0.72	~1.00	What fraction of true gullies are detected
ROC-AUC	> 0.90	~0.50	Discrimination ability across all thresholds

10. Outputs, Exports and the Dashboard

10.1 Cloud-Optimized GeoTIFF (COG)

All probability maps and binary masks are exported as Cloud-Optimized GeoTIFFs with DEFLATE compression, internal tiling (256x256 blocks), and pyramid overviews (2x, 4x, 8x, 16x). COGs can be streamed over HTTP range requests — a GIS user in another institution can load the result directly from a URL without downloading the full 2+ GB file, enabling efficient scientific data sharing.

10.2 STAC 1.0 Catalog

All outputs are registered in a STAC (SpatioTemporal Asset Catalog) 1.0 catalog. STAC is the emerging open standard for geospatial data cataloging, supported by ESA, NASA, and Copernicus. The catalog includes one STAC Item per observation epoch, with assets linking to the COG probability maps, binary masks, and change products. This enables integration with tools like QGIS, Python's pystac, and web-based STAC browsers.

10.3 Interactive Folium Dashboard

The dashboard (`data/outputs/dashboard/mars_gully_dashboard.html`) is a self-contained HTML file (~5-15 MB) with:

- **Mars basemaps:** THEMIS IR Day, MOLA colour-coded DEM, CTX Murray Lab mosaic — via WMS tile layers
- **Probability overlay:** Latest-epoch probability map with blue-red colormap, opacity 0.7
- **Binary mask overlay:** `thr=0.5` mask in orange, selectable by date
- **Change layer:** Gain (green), stable (yellow), loss (red) composite
- **GeoJSON polygons:** Clickable gully polygons with area, date, expansion rate metadata
- **Activity markers:** Per-tracked-gully CircleMarkers with popup showing Kalman-smoothed history
- **Time-series panel:** Embedded Plotly chart showing Kalman-smoothed area with uncertainty band
- **Layer control:** Toggle each layer on/off; basemap selector

10.4 Summary Reports (CSV)

Report file	Columns	Purpose
<code>per_site_summary.csv</code>	<code>site_id</code> , <code>n_gully_pixels</code> , <code>gully_area_ha</code> , <code>n_polygons</code> , <code>prob_mean</code> , <code>prob_max</code>	Site-level summary per epoch
<code>change_report.csv</code>	<code>date_t0</code> , <code>date_t1</code> , <code>gain_ha</code> , <code>loss_ha</code> , <code>stable_ha</code> , <code>net_change_ha</code> , <code>kalman_velocity_ha_year</code>	Inter-epoch change statistics
<code>activity_report.csv</code>	<code>gully_id</code> , <code>first_seen</code> , <code>last_seen</code> , <code>area_ha_latest</code> , <code>total_gain_ha</code> , <code>expansion_rate</code>	Per-tracked-gully activity record

11. Running the Pipeline — Quick Reference

Installation

```
# Clone and set up environment
git clone https://github.com/your-username/mars-gully-digital-twin
cd mars-gully-digital-twin

# Option A: Docker (recommended)
docker build -t mars-gully .
docker run --gpus all -v $(pwd)/data:/workspace/data mars-gully \
python main.py --step all

# Option B: Local Python (Windows)
conda create -n mars-gully python=3.10 -y
conda activate mars-gully
pip install -r requirements.txt
```

Step Commands

Command	What It Does	Output Location
python main.py --step download	Downloads HiRISE, CTX, MOLA data	data/raw/
python main.py --step preprocess	Coregister, project, normalise all imagery	data/processed/
python main.py --step features	Build 8-channel feature stacks	data/processed/feature_stacks/
python main.py --step labels	Generate synthetic gully masks from MOLA	data/processed/labels/
python main.py --step train	Train U-Net (100 epochs, early stop)	data/models/unet_best.pth
python main.py --step train --resume last	Resume interrupted training	data/models/
python main.py --step infer	Sliding-window inference on all stacks	data/outputs/probability_maps/
python main.py --step threshold	Apply thresholds, clean small components	data/outputs/binary_maps/
python main.py --step change	Change detection + Kalman + tracking	data/outputs/change_detection/
python main.py --step export	COG, GeoJSON, CSV, STAC catalog	data/outputs/stac/reports/
python main.py --step dashboard	Build interactive HTML dashboard	data/outputs/dashboard/
python main.py --step validate	Spatial block cross-validation	data/outputs/reports/spatial_cv.json

<code>python main.py --step validate --temporal</code>	Temporal hold-out (2006-10 vs 2015-20)	<code>data/outputs/reports/temporal_cv.json</code>
<code>python main.py --step all</code>	Full pipeline end-to-end	All of the above

Resume and Restart

Every expensive step writes a JSON checkpoint file. If a step is interrupted, re-running the same command continues from the last completed sub-step — no data is re-downloaded or re-processed. Checkpoint files are stored in `data/outputs/_checkpoint.json`.

```
# Check what has been completed
cat data/outputs/inference_checkpoint.json

# Force rerun a completed step
python main.py --step infer # (add force=True in config or delete checkpoint)

# Run tests
pytest tests/ -v --tb=short
```

12. Expected Results and Publication Targets

Quantitative Performance Targets

Based on comparable segmentation tasks on Mars HiRISE imagery (Wagstaff et al. 2018; Palafox et al. 2017) and the richness of our 8-channel feature stack, we project:

Metric	Conservative	Expected	Optimistic
Val IoU (spatial CV)	0.55	0.65	0.72
Val F1 (spatial CV)	0.65	0.75	0.82
Temporal hold-out IoU	0.48	0.58	0.67
Generalisation gap	< 0.12	< 0.08	< 0.05
Change area precision	0.65	0.75	0.82

Scientific Expected Findings

- **Gasa Crater:** Measurable area gain of 0.5-3.0 ha over the 2007-2018 observation window, consistent with 2-4 new deposit events identified by Dundas et al.
- **Palikir Crater:** Seasonal RSL extent correlated with Ls (solar longitude), shrinking in winter and expanding in summer — providing a validation against published RSL measurements.
- **Russell Crater:** Linear gully growth rate of 0.1-0.5 ha/year on dune slip faces, consistent with CO₂ sublimation model predictions (Pilorget & Forget 2016).
- **Cross-site generalisation:** Model trained on Gasa generalises to Russell with IoU drop < 0.10 — evidence that common spectral-morphometric signatures underlie gully formation across sites.

Target Publications

Journal	Scope	Target Section
Icarus (Elsevier)	Primary planetary science journal	Mars: Surface/geology; expected 2026-2027
Earth and Space Science (AGU)	Methods + Earth/Mars comparisons	Geospatial ML applications
Remote Sensing (MDPI)	Methods-focused open access	Deep learning for planetary remote sensing
Journal of Geophysical Research: Planets	High-impact planetary geomorphology	Gully change quantification results

Relevance to Copernicus Master's Research Themes

This project directly addresses three of the four Copernicus Master's research tracks: **Digital Earth** (STAC catalogs, COG, interoperable data pipelines), **Remote Sensing and Image Analysis** (multi-sensor fusion, deep learning segmentation, change detection), and **Planetary Observation** (Mars orbital data processing, PDS archive navigation, planetary CRS handling). The methodological contributions — spatial block CV, domain-adapted attention U-Net, Kalman-smoothed area tracking — are directly transferable to

Earth observation gully and landslide monitoring pipelines.

The Mars Gully Digital Twin is simultaneously a scientific instrument for planetary geomorphology research AND a demonstration of production-grade Earth Observation data engineering — exactly the dual mandate of the Copernicus Master's in Digital Earth programme.

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Pipeline source: github.com/your-username/mars-gully-digital-twin

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